

APPLICATION

estar: An R package to measure ecological stability

Ludmilla Figueiredo^{1,2}  | Cédric Scherer¹ | Stephanie Kramer-Schadt^{1,3}  |
 Juliano Sarmento Cabral^{2,4,5} | Sonia Kéfi^{6,7} | Paul J. Van den Brink⁸ | Viktoriia Radchuk¹ 

¹Department of Ecological Dynamics, Leibniz Institute for Zoo and Wildlife Research (IZW), Berlin, Germany; ²Ecosystem Modeling Group, Center for Computational and Theoretical Biology (CCTB), University of Würzburg, Würzburg, Germany; ³Institute of Ecology, Technische Universität Berlin, Berlin, Germany; ⁴School of Biosciences, College of Life and Environmental Sciences, University of Birmingham, Birmingham, UK; ⁵Ecological Modelling Group, Department of Plant Biodiversity, Bonner Institute for Organismal Biology, University of Bonn, Bonn, Germany; ⁶ISEM, CNRS, Univ. Montpellier, IRD, EPHE, Montpellier, France; ⁷Santa Fe Institute, Santa Fe, New Mexico, USA and ⁸Aquatic Ecology and Water Quality Management Group, Wageningen University, Wageningen, The Netherlands

Correspondence

Viktoriia Radchuk

Email: radchuk@izw-berlin.de**Funding information**

Deutsche Forschungsgemeinschaft,
 Grant/Award Number: RTG 2118-1;
 Alexander von Humboldt Foundation

Handling Editor: Luis Cayuela**Abstract**

1. Assessing ecological stability is a key task in biodiversity monitoring and conservation. Quantifying stability is not trivial because its different aspects can be measured with various properties. However, to date, no software enables measuring different stability properties on ecological time-series data.
2. We present the `estar` R package that standardises and facilitates the use of 11 established stability properties that have been used to assess systems' responses to press or pulse disturbances at different ecological levels (e.g. population, community).
3. `estar` provides two sets of functions. The first set corresponds to functions that measure stability at any level of organisation, from individual to community, and can be applied to time series of a system's state variables (e.g. body mass, population abundance or species diversity). The properties included in this set are: invariability, resistance, extent and rate of recovery, persistence and overall ecological vulnerability. The second set of functions can be applied to Jacobian matrices. The functions in this set measure the stability of a community at short and long time scales. In the short term, the community's response is measured by maximal amplification, reactivity and initial resilience (i.e. initial rate of return to equilibrium). In the long term, stability can be measured as asymptotic resilience and intrinsic stochastic invariability.
4. The package includes vignettes demonstrating the use of all functions and an introduction to the multivariate autoregressive state-space models necessary for the second set of functions. `estar` constitutes a toolbox with standardised, ready-to-use functions that bridge dichotomies in definitions and enable comparisons across state variables, taxa and scales.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Methods in Ecology and Evolution* published by John Wiley & Sons Ltd on behalf of British Ecological Society.

KEYWORDS

community stability, first-order multivariate autoregressive state-space model, population stability, species interaction matrix

1 | INTRODUCTION

Diverse anthropogenic disturbances such as habitat degradation and destruction, pollution and species invasions affect stability at different levels of organisation, from individuals to populations and communities. Indeed, disturbances affect physiological and phenotypic traits of individuals, which may lead to changes in population dynamics and stability of single species and reverberate in community stability and functioning of the whole ecosystem (Figure 1). Therefore, measuring stability at different levels of ecological organisation is of utmost importance for biodiversity monitoring and conservation.

Stability is broadly defined as the ability of the system to retain its function and structure in the face of disturbances (Holling, 1973). Since its introduction in the 1950s, the concept of stability has been shown to have a multidimensional character (Donohue et al., 2013; Grimm & Wissel, 1997; Pimm, 1984; Van Meerbeek et al., 2021) that manifests itself in at least three ways. First, different stability properties such as resistance, recovery and invariability are conceptually independent in that they measure different aspects of stability (Domínguez-García et al., 2019;

Downing et al., 2020; Neubert & Caswell, 1997; Radchuk et al., 2019), making it necessary to quantify several of them when studying a system's stability (but see Urrutia-Cordero et al., 2022). Secondly, the same stability property can be measured at different organisational levels, for example, stability of community composition and ecosystem function are referred to as 'compositional' and 'functional stability', respectively (Hillebrand & Kunze, 2020; Urrutia-Cordero et al., 2022). Similarly, the same stability property can be calculated using different *state variables* (Table 1). For example, compositional diversity can be quantified using one of the following state variables: a diversity metric (e.g. the Shannon index or species richness), community abundance and community dissimilarity (e.g. Bray–Curtis distance) to the undisturbed community. Thirdly, different metrics have been proposed to calculate a single stability property (we henceforth use 'metrics' to refer to the quantification(s) of a property implemented in *estor*). For example, resistance is often measured immediately after a disturbance but can also be measured when the difference between the state variable of the disturbed system and its pre-disturbed state is the largest (White et al., 2020; also known as 'maximum attenuation', Capdevila et al., 2020). These issues hamper synthesis

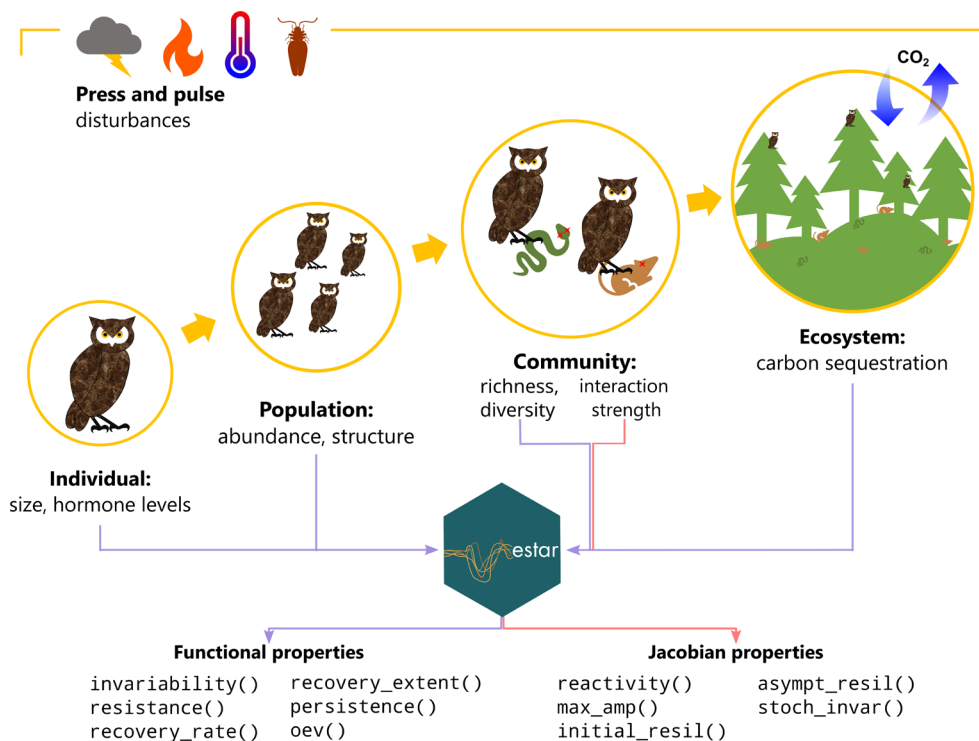


FIGURE 1 The *estor* package can measure stability at different ecological levels in response to press and pulse disturbances, by taking in state variables at different levels of organisation: Individual, population, community and ecosystem. The functional metrics can be calculated for all levels (purple arrows), while the Jacobian metrics can only be calculated for the community, based on the matrix of species interactions (orange arrows).

TABLE 1 Glossary.

Baseline: state variable value(s) the user defines to represent an undisturbed system. Also known as 'reference state'. It can be measured before a disturbance affects the focal system, from a separate undisturbed system or from a single point considered to be representative of an undisturbed system (Figure 2).

Jacobian matrix: A functional matrix or derivative matrix of a differentiable function of all first partial derivatives in the case of total differentiability. The Jacobian is used, for example, to approximate multidimensional functions in mathematics or, in the context of community ecology, for functions of interaction strength. See *Species interaction matrix* in this table.

Press disturbances: A disturbance that affects a system continuously for an extended period (Ryo et al., 2019), for example, climate change or ongoing pollution from leakage.

Pulse disturbance: An event that suddenly affects a system and recedes quickly after reaching a peak, for example, fires or storms (Ryo et al., 2019).

Species interaction matrix (B): A matrix quantifying the strength of density dependence between the n species in a community, that is, the effect of one species' density on the per capita growth rate of another (or of its own, Hampton et al., 2013). It is often referred to as 'community matrix' (or 'a discrete-time version of a *Jacobian matrix*'; Downing et al., 2020).

State variable: A variable describing a system, for example, population abundance (whereby the population is a system) or species richness (whereby the community is a system).

across ecological stability studies. Beyond the necessity of quantifying multiple properties of stability, there is one practical concern: No software permits quantifying stability in the diverse ways it has been measured so far. Therefore, the time is ripe for a tool to measure different stability properties (or dimensions) in a standardised, comparable and reproducible way.

In addition to the multifaceted nature of the stability concept, there is a large divide between the properties used by empiricists and theoreticians (Donohue et al., 2016). While empirical studies often measure the temporal invariability of population and community state variables, theoretical studies mainly quantify asymptotic stability properties derived from species interaction matrices (*species interaction matrix*, Table 1, Donohue et al., 2016), which are rarely available for observational studies. Assessing interaction strength in empirical systems is a rather laborious task and can be accomplished with various methods (e.g. controlled pairwise species experiments; Carrara et al., 2015a). However, estimations from different methods are usually not comparable (Carrara et al., 2015b). In that context, Multivariate Autoregressive Models (MARs) are a flexible method, as they can be applied to time-series data on community composition to derive species interaction matrices, which can be used to derive asymptotic stability properties (Downing et al., 2020; Ives et al., 2003). Still, the use of MARs outside freshwater plankton community studies remains limited (Hampton et al., 2013). Therefore, a user-friendly tool that integrates MARs and allows

the user to derive asymptotic stability properties has the potential to close the gap between empirical and theoretical stability research.

Here, we present *estar*, an R package designed to facilitate the use of 11 stability properties. We distinguish two groups of properties (Figure 1). We refer to one group as 'functional properties' in a very broad sense, encompassing what has been previously referred to in the literature as 'functional' and 'compositional' properties (Hillebrand et al., 2018; Hillebrand & Kunze, 2020; Urrutia-Cordero et al., 2022), and extending it to include stability at the individual level. These properties measure the stability of functioning at different levels of organisation: in the same way that the functioning of single organisms can be assessed through phenotypic traits, population functioning through population dynamics and ecosystem functioning through ecosystem functions, community functioning can be assessed through changes in its composition. This group encompasses the following properties: invariability, resistance, extent of recovery, rate of recovery, persistence and overall ecological vulnerability (Table 2). We refer to the second group as 'Jacobian properties' because they are derived from the species interaction matrix (also referred to as 'Jacobian matrix'). Jacobian properties quantify the stability of community composition only, unlike functional properties. The input for computing Jacobian properties is a matrix of species abundances over time or a species interaction matrix. The following properties belong to this group: maximal amplification, initial resilience, asymptotic resilience, intrinsic stochastic invariability and intrinsic deterministic invariability (Table 3). We showcase the use of the package to measure the stability of a real-world freshwater invertebrate community perturbed by an insecticide (van den Brink et al., 1996).

2 | AN OVERVIEW OF THE EXISTING SOFTWARE TO QUANTIFY STABILITY

Although ecological stability attracts much research interest, we still lack R and Python software to readily calculate the variety of stability properties defined in the literature. At the community level, the *codyn* package (Hallett et al., 2020) provides four functions to measure stability in terms of species variances and covariance. The package *MAR1* (Scheef & Holmes, 2023) offered functionality to both fit MARs and derive asymptotic stability properties from the output of those models, but it has not been maintained since 2019. The *MARSS* package, which can be used to estimate species interaction matrices (Table 1), does not provide formulas for calculating stability properties (sensu Ives et al., 2003). The R package *popdemo* (Stott et al., 2021) focuses on the population level and allows calculating demographic resilience using transient analyses (Capdevila et al., 2020). Furthermore, although many stability properties (e.g. resistance and recovery rate) apply to a single time series (univariate data) and seem conceptually simple, no single software package allows quantifying all of them simultaneously. Thus, a tool is needed

TABLE 2 Definition, application and interpretation of the variants of the functional stability metrics. For each metric, we specify the input data and time points required for its calculation, as well as possible responses (e.g. log response ratio or dissimilarity). Definitions are given separately for compositional stability (calculated as the dissimilarity between the disturbed and baseline communities) and functional stability (calculated from a time series of a state variable at any level of organisation). v_b refers to an independent baseline; v_p , to a baseline defined by pre-disturbance values, which can be summarised as mean or median for some metrics; and v_d , to the state variable values in the disturbed time series; t stands for time.

Metric and application	Formal definition(s)	Notation	Interpretation
Invariability (I); for pulse disturbance, calculated starting from the time step when resistance is measured until recovery; for press or environmental stochasticity, calculated over the whole time the disturbance is applied (see also Urrutia-Cordero et al., 2022).	The inverse of the coefficient of variation of the response. For functional stability, the response is the log response ratio between the state variable's value in the disturbed and in the baseline systems or the state variable's value in the disturbed system itself. For compositional stability, it is the dissimilarity between the disturbed and baseline communities The inverse of the standard deviation of residuals of the linear model where the response (same as above) is predicted by time	$I = CV\left(\log\left(\frac{v_d}{v_b}\right)\right)^{-1}, I = CV\left(\log\left(\frac{v_d}{v_b}\right)\right)^{-1}$ $I = CV(v_d)^{-1}$ $I = CV\left(\text{dissim}\left(\frac{C_d}{C_b}\right)\right)^{-1}$ $I = \sigma(\epsilon)^{-1}, \text{ from } y = \alpha + R_r t + \epsilon, \text{ where } y \in \left\{\log\left(\frac{v_d}{v_b}\right), \log\left(\frac{v_d}{v_p}\right), v_d, \text{dissim}\left(\frac{C_d}{C_b}\right)\right\} \text{ and } \alpha \text{ is the intercept}$	The larger in magnitude I is, the higher the stability, since the variation around the trend is lower
Resistance (R); for pulse only	For functional stability, the log response ratio or absolute difference between the state variable's value in the disturbed and in the baseline systems, on the user-defined time step^a. For compositional stability, the dissimilarity between disturbed and baseline communities For functional stability, the maximal log response ratio or absolute difference between the state variable's value in the disturbed and the baseline systems, over a user-defined time interval. For compositional stability, the maximal dissimilarity	$R = \log\left(\frac{v_d(t)}{v_b(t)}\right), R = \log\left(\frac{v_d(t)}{v_p(t)}\right)$ $R = v_d(t) - v_b(t) , R = v_d(t) - v_p(t) $ $R = \text{dissim}\left(\frac{C_d(t)}{C_b(t)}\right)$ $R = \max_t\left(\log\left(\frac{v_d(t)}{v_b(t)}\right)\right), R = \max_t\left(\log\left(\frac{v_d(t)}{v_p(t)}\right)\right),$ $R = \max_t(v_d(t) - v_b(t)), R = \max_t(v_d(t) - v_p(t)), \text{ or}$ $R = \max_t\left(\text{dissim}\left(\frac{C_d(t)}{C_b(t)}\right)\right)$	Maximum resistance at 0. Smaller or higher values indicate under- or overcompensation, respectively For Bray–Curtis index, $0 \leq R \leq 1$. Maximum resistance at 0. The higher the index, the more apart the communities and thus lower resistance Same interpretation as above
Extent of recovery (R_e); press and pulse	For functional stability, the log-response ratio or difference between the state variable in the disturbed system and the baseline measured on the user-defined time step t_{post} when the recovery is assumed to have taken place^b. For community stability, the dissimilarity between disturbed and baseline communities at t_{post}	$R_e = \log\left(\frac{v_d(t)}{v_b(t)}\right), R_e = \log\left(\frac{v_d(t)}{v_p(t)}\right), \text{ with } t = t_{\text{post}}$ $R_e = v_d(t) - v_b(t) , R_e = v_d(t) - v_p(t) , \text{ with } t = t_{\text{post}}$ $R_e = \text{dissim}\left(\frac{C_d(t)}{C_b(t)}\right), \text{ with } t = t_{\text{post}}$	Maximum recovery at 0. Smaller or higher values indicate under- or overcompensation, respectively For Bray–Curtis index (default), $0 \leq R_e \leq 1$. The higher the index, the further apart the communities are after recovery
Rate of recovery (R_r); press and pulse	The slope of a linear model, where the response is predicted by time. The responses can be the same as the ones used to calculate invariability	$R_r = \frac{\sum(t - \bar{t})(y - \bar{y})}{\sum(t - \bar{t})^2}, \text{ where}$ $y \in \left\{v_d, \log\left(\frac{v_d}{v_b}\right), \log\left(\frac{v_d}{v_p}\right), v_d, \text{dissim}\left(\frac{C_d}{C_b}\right)\right\}$	If $R_r = 0$, the system did not react to the disturbance. For functional stability, if $R_r > 0$, the system moved towards the values in the baseline after the disturbance (recovery, which may be partial). If $R_r < 0$, the system deviated even further from the control. For compositional stability, the opposite is true. In both cases, the higher R_r, the faster the response

(Continues)

TABLE 2 (Continued)

Metric and application	Formal definition(s)	Notation	Interpretation
Persistence (P); pulse and press (but harder to compare in press, see Pimm, 1984)	For functional stability, the proportion of the user-defined time frame t_a during which the response stayed within the limits of an interval determined by the baseline mean $\pm$ SD (t_p). For functional stability, the baseline is v_b or v_p , and the response, v_d . For community stability, the proportion of the user-defined time frame t_a during which community dissimilarity remained within user-defined upper and lower limits of the community dissimilarity	$P = t_p / t_a$	$0 \leq P \leq 1$. The higher the persistence is, the more stable the system is
Overall ecological vulnerability (OEV), press and pulse	The area under the curve (AUC) of the system response through time (Urrutia-Cordero et al., 2022). The response can be the absolute log-response-ratio (functional stability) or the dissimilarity (compositional stability) between the disturbed and baseline communities	$OEV = AUC \left(\left \log \left(\frac{v_d(t)}{v_b(t)} \right) \right , t \right)$, $OEV = AUC \left(\left \log \left(\frac{v_d(t)}{v_p(t)} \right) \right , t \right)$, $OEV = AUC \left(\text{dissim} \left(\frac{C_d(t)}{C_b(t)} \right), t \right)$, where $AUC(y, t) = \sum_{i=1}^{n-1} \frac{(t_{i+1} - t_i)(y_i - y_{i+1})}{2}$	$OEV \geq 0$. The higher the value, the less stable the system

Note: a usually the first time step following disturbance and b usually the last time step of the time series. The timesteps and different inputs each function can use are illustrated in Figure 2.

TABLE 3 Definitions of the Jacobian stability metrics, based on species interactions matrix ($\mathbf{B}$).

Metric, function name	Formal definition and interpretation	Equation
Reactivity (R_a , <code>reactivity()</code>)	Maximum instantaneous rate at which a perturbation can grow at the moment of disturbance (Neubert & Caswell, 1997). If $R_a > 0$, the perturbation grows, that is, the system is initially destabilised. If $R_a < 0$, the perturbation does not grow and the system is not initially destabilised	$R_a = \lambda_{\text{dom}}(H(\mathbf{B}))$, where λ_{dom} is the dominant eigenvalue, and H is the Rayleigh quotient
Maximal amplification (A_{max} , <code>max_amp()</code>)	Largest factor by which a perturbation is amplified (relative to its initial size) before the system's eventual equilibrium. If $A_{\text{max}} > 1$, the system overreacts and departs from equilibrium (it might stabilise after the initial transient period, but A_{max} does not access that). If $A_{\text{max}} \approx 1$, the system is stable. It is calculated as the Euclidean norm of the species interaction matrix (Neubert & Caswell, 1997)	$A_{\text{max}} = \max_{t \geq 0} \left(\max_{x_0 \neq 0} \frac{\ e^{\mathbf{B}^T t} x_0\ }{\ x_0\ } \right)$, where $\ e^{\mathbf{B}^T t}\ $ is the matrix norm of $\mathbf{B}$ and x_0 is the vector of initial abundances (Domínguez-García et al., 2019)
Initial resilience (R_0 , <code>init_resil()</code>)	Initial resilience is calculated as the initial rate of return to equilibrium (Downing et al., 2020). The larger its value, the more stable the system, as its 'worst case' initial rate of return to equilibrium is faster (Downing et al., 2020)	$R_0 = -\log \left(\sqrt{\lambda_{\text{dom}}(\mathbf{B}^T \mathbf{B})} \right)$
Asymptotic resilience (R_{∞} , <code>asym_resil()</code>)	The slowest/long-term asymptotic rate of return to equilibrium after a pulse perturbation (Arnoldi et al., 2016; Downing et al., 2020). R_{∞} is a positive real number. The larger its value, the more stable the system, as its return to equilibrium is faster (Downing et al., 2020)	$R_{\infty} = -\log(\lambda_{\text{dom}}(\mathbf{B}))$
Intrinsic stochastic invariability (I_s , <code>stoch_var()</code>)	Inverse of the maximal response variance to white noise (Arnoldi et al., 2016). The larger its value, the more stable the system	$I_s = \frac{1}{\ \mathbf{B}^{-1}\ }$ where $\ \mathbf{B}^{-1}\ $ is the spectral norm of the inverse of matrix $\mathbf{B}^{-1} = \mathbf{B} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{B}$, with $\mathbf{I}$ being an identity matrix

to derive both asymptotic stability properties at the community level and stability properties from time-series data applicable any level of organisation. To address this, we provide the *estar* package.

3 | PACKAGE OVERVIEW

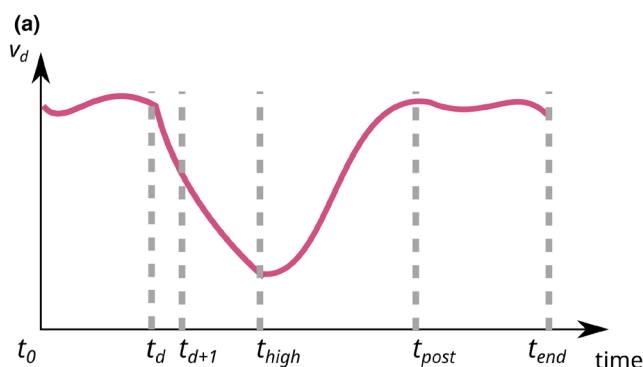
estar is available from CRAN and can be installed via `install.packages('estar')` for $R \geq 4.1$ (Figueiredo et al., 2025). After installation, all functions are ready for use. The main functions calculate the functional and Jacobian metrics (Figure 1), and each of them outputs a single value for the stability metric it is named after (Tables 2 and 3; other functions are for internal use and not described here). The functions for functional metrics require data organised in wide-format data frames, where each variable occupies one column (i.e. time, state variable in disturbed/undisturbed systems, and species abundances or presence/absence for compositional stability). Since *baseline* (Table 1) is a central concept for quantifying stability

(Ingrisch & Bahn, 2018), *estar* gives the user flexibility to specify whether the state variable should be normalised by the baseline (as the log response ratio or difference, Figure 2c,d). The user can choose the data used as baseline: data prior to the disturbance from the disturbed system (a time frame or point) or the data from an independent undisturbed system (Figure 2b). The data frame (or data frames, if the baseline comes from an undisturbed system) is given as input, along with relevant variable names, and argument values defining which variant of a metric should be calculated (Table 2).

4 | FUNCTIONAL STABILITY PROPERTIES

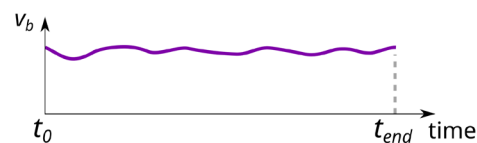
At each level of organisation, functional stability properties can be calculated using a time series of a relevant state variable as input (Figures 1–3, Table 2). In addition, compositional stability can be calculated, as is commonly done in community ecology, from the dissimilarity between the disturbed and baseline communities. For this,

Minimal data input

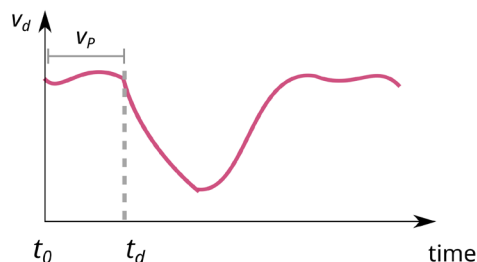


Possible baseline time-series

(b1) Undisturbed time-series

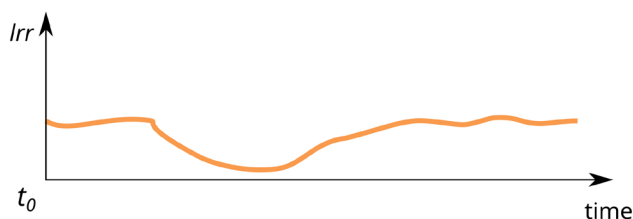


(b2) Pre-disturbance time-series



Data transformation (wrapped in the package functions)

(c) Log response ratio ($lrr = \log(v_d/v_b)$)



(d) Difference ($diff = v_d - v_b$)

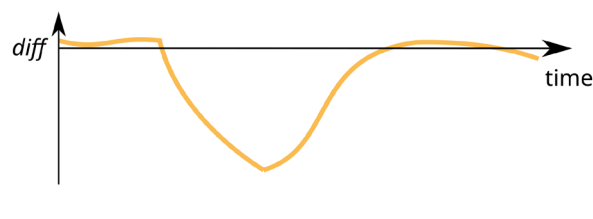


FIGURE 2 Possible univariate inputs to functional stability properties (a, b) and the two types of transformation applied to them by some metrics (c, d). Relevant time steps: t_d , time when pulse disturbance is applied to the system; t_{d+1} , first time step after disturbance, t_{high} time step where the absolute distance between state variable value in the disturbed system and baseline is the highest, t_{post} time step where recovery is considered to have happened, t_{end} end of time series. (a) The data for which a metric is to be calculated must constitute a time series of a state variable in the disturbed system (v_d). (b) Most functions require a time series of the same state variable in the baseline system. This baseline can be taken from a separate undisturbed system (v_b , b.1) or from the pre-disturbance values of the disturbed system (v_p , b.2). It can be a single value or a summary of the time series. The stability metrics that require both the disturbed and baseline time series are applied to either the log response ratio (*lrr*) between the values in these two time series (c) or the difference between them (d).

we rely on the `vegdist` function from the package `vegan` (Oksanen et al., 2024), whose parameters are inherited by `estar`, allowing it to work with a variety of distances. Importantly, the calculation of the functional stability metrics requires user-defined time points or periods (Table 2). These do not have a default value since they are system-dependent and must be provided by the user who knows their system best (or is willing to explore the effects of such a choice on the metric).

Invariability (I , `invariability()`, Figure 3a), the most common stability measure (Donohue et al., 2016), measures the system's overall variation under pulse and press disturbances, as well as environmental stochasticity. **Resistance** (R , `resistance()`, Figure 3b) is the magnitude of change in the state variable following a pulse disturbance (Pimm, 1984). It can be calculated as the maximum magnitude of change in the state variable or the magnitude at a user-defined time step after the disturbance (e.g. at the first time step to capture the system's initial resistance). **Recovery** is the return of the system's state variable to the baseline state (Medeiros et al., 2021; Van Meerbeek et al., 2021). `estar` contains two properties for it: the **extent of recovery**, that is, how close the system returns to the baseline (E_r , `recovery_extent()`, Figure 3c) and the **rate of recovery**, that is, how fast it returns to the baseline (R_r , `recovery_rate()`, Figure 3d). The rate of recovery in the literature is often referred to as 'resilience' (Hillebrand et al., 2018; Pimm, 1984; Urrutia-Cordero et al., 2022). **Persistence** (P , `persistence()`, Figure 3e) is the proportion of time a variable stays

within one standard deviation from the baseline's mean during the user-defined period (based on Pimm, 1984). Finally, **overall ecological vulnerability** (OE_V , `oev()`, Figure 3f) integrates the other stability metrics and reflects the area under the curve of the system's response over time (Urrutia-Cordero et al., 2022). The vignette 'Functional stability properties' contains a complete description of the variants of each metric and an evaluation of the functions' performance.

5 | JACOBIAN STABILITY PROPERTIES

The Jacobian properties measure the community's responses to a disturbance, both in the short- and in the long term. The long-term (asymptotic) rates are typically estimated for theoretical systems (Arnoldi et al., 2018; Neubert & Caswell, 1997), whereas empirical studies mainly assess the short-term (transient) rates (Arnoldi et al., 2018). The long-term data are rarely available for empirical systems, limiting the possibility of deriving dominant eigenvalues from *Jacobian matrices* (Table 1) to calculate the asymptotic resilience of theoretical systems (Table 3).

Jacobian properties are calculated from species interaction matrices (denoted as B , Table 1). Our package allows the estimation of B from the user-supplied time series data on species abundances in the community. We provide the function (`extractB()`) to format the output of the multivariate autoregressive state space (MARSS)

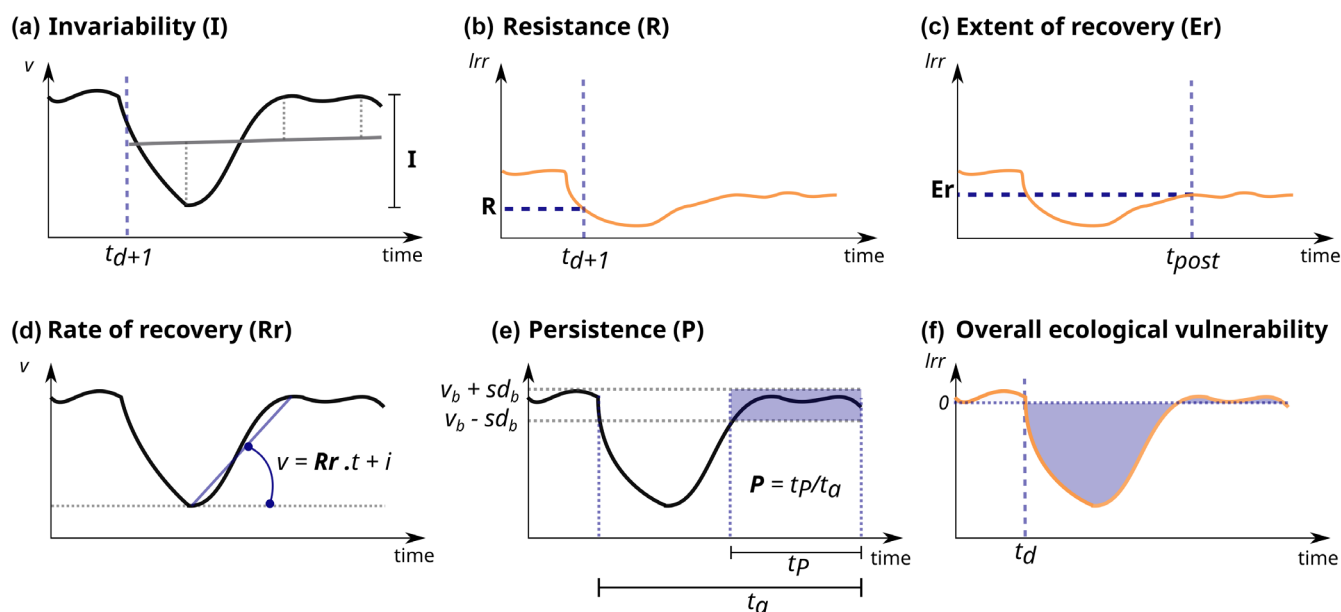


FIGURE 3 Variants of functional stability properties calculated by `estar`: (a) invariability (I) as the coefficient of variation of the state variable in the disturbed system; (b) resistance (R) as the log-ratio (lrr) between the disturbed and baseline time series in the first time step after disturbance (t_{d+1}); (c) extent of recovery (E_r) as log response ratio between the disturbed and baseline time series at the user-defined time step the system is expected to have recovered (t_{post}); (d) rate of recovery (R_r) as the slope of the linear model fitted to the disturbed time series; (e) persistence (P) calculated as the proportion of time during which the disturbed time series stayed in the interval defined as $\pm 1SD$ from the baseline ($v_b \pm sd_b$) over the total user-specified time period (t_a); and (f) overall ecological vulnerability (OE_V) as the area under the curve of the log response ratio between the disturbed and baseline time series. Depending on the metric, it can be calculated from a time series of a state variable (v ; black line) and/or the log response ratio (lrr ; orange line). All variants are demonstrated in the 'Functional stability properties' vignette.

model fitted by the R package MARSS (Holmes et al., 2012; Holmes, Scheuerell, & Ward, 2024; Holmes, Ward, et al., 2024). MARSS models estimate $\mathbf{B}$ while accounting for uncertainty in the observation process that generated the data. We exemplify how MARSS models work and comment on possible pitfalls and the functions' performance in the 'Jacobian stability properties' vignette.

We provide three functions to characterise a community's transient response to a pulse disturbance (Figure 4). Two of these functions calculate the metrics describing the 'amplification envelope'—the curve describing the upper bound response to a disturbance (Neubert & Caswell, 1997). The amplification envelope can be interpreted as an estimation of initial instability (Arnoldi et al., 2016) and is characterised by measures of *reactivity* (R_a , `reactivity()`, Table 3) and *maximal amplification* ($A_{\max}$, `max_amp()`). Complementing the amplification envelope, the *initial resilience* (R_0 , `init_resil()`) characterises the system's initial rate of return to equilibrium (Table 3, Arnoldi et al., 2018).

To measure stability in the long term, we provide a function to calculate *asymptotic resilience* (R_∞ , `asympt_resil()`, Table 3, Figure 4), the long-term rate of return to equilibrium, complementary to R_0 (Arnoldi et al., 2018). Further, to measure instability under press disturbance or white noise, *estar* provides the *intrinsic stochastic invariability* (I_∞ , `stoch_var()`), a theoretical equivalent of the functional metric 'invariability' (Arnoldi et al., 2016), albeit uncorrelated to it (Downing et al., 2020).

6 | EXAMPLE: STABILITY OF AN AQUATIC MACROINVERTEBRATE COMMUNITY

To exemplify the use of *estar*, we analysed the stability of an experimental community of freshwater macroinvertebrates disturbed

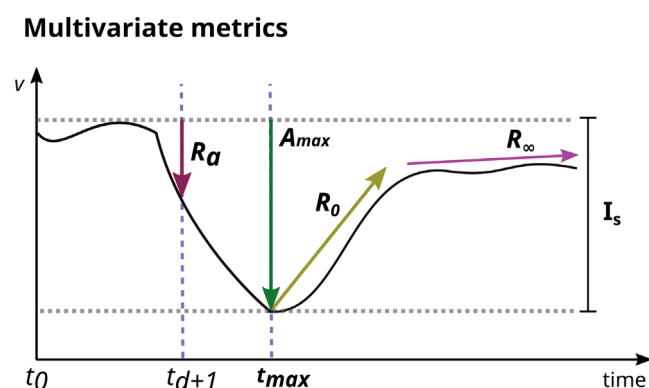


FIGURE 4 Schematic representation of the Jacobian stability properties: Reactivity (R_a), calculated at the first time step following disturbance (t_{d+1}); maximal amplification ($A_{\max}$); initial resilience (R_0); asymptotic resilience (R_∞); and intrinsic stochastic invariability (I_s). To facilitate comprehension, we illustrate metrics in relation to a state variable (v) disturbed at time t_d , but the metrics, in fact, are calculated from the community's species interactions matrix and not a single state variable. All properties are demonstrated in the 'Jacobian stability metrics' vignette.

by the insecticide chlorpyrifos (van den Brink et al., 1996, Figure 5a). Species in this community were distinguished into three functional groups: carnivores, detritivores and herbivores. We analysed the effect of two concentrations of the insecticide on the functional and Jacobian stability properties (Figure 5b,c, Table 4 and Tables S1–S4, fully described in Data S1). We calculated functional properties using three state variables: community abundance, diversity as Shannon–Wiener index and composition as Bray–Curtis dissimilarity compared to the control treatment.

Functional properties calculated with different state variables (abundance, diversity and composition) show a variety of behaviours for the same stability metric. For example, the overall ecological vulnerability is higher under high insecticide concentration—and thus stability is lower—when calculated using diversity, but when using other state variables, it does not distinguishably differ between insecticide treatments, that is, confidence bars overlap. When analysing each group's abundance, detritivores recovered better than carnivores and herbivores under the high insecticide concentration, as quantified by the extent of recovery and rate of recovery (at week 28), despite having lower resistance. At the community level, diversity and composition show different behaviours: higher overall ecological vulnerability for diversity (Bray–Curtis) at high concentration, but no distinguished response in terms of composition (Shannon index). Regarding the Jacobian properties (Figure 5c), the higher reactivity and maximal amplification values at high insecticide concentration reflect the strong decreases in herbivore and detritivore abundances. We found mixed evidence for relations between Jacobian and functional stability properties. On the one hand, Jacobian reactivity was higher in the high insecticide treatment, which corresponded to its functional counterpart, rate of recovery, measured with composition, being lower in the high treatment. However, rate of recovery measured with diversity as a state variable does not suggest a difference between the treatments. On the other hand, the results of maximal amplification are mirrored by resistance quantified both as diversity and as composition (Figure 5b,c). Overall, a variety of responses assessed using different state variables and different metrics highlight the importance of a comprehensive analysis of stability and underscore the utility and timeliness of *estar*.

7 | CONCLUSION

The *estar* package provides functions for calculating stability properties, grouped in two sets: (1) functions to quantify stability at different levels of organisation, from individual to community (functional stability properties) and (2) functions to quantify stability from species interaction strengths (Jacobian stability properties). With the first set, *estar* allows the quantification of stability in a comparable manner across organisational levels and systems, which we hope will contribute to reconciling the dichotomies in definitions in stability research. With the second set, the package comes out as the first tool for easy computing of the stability properties that were

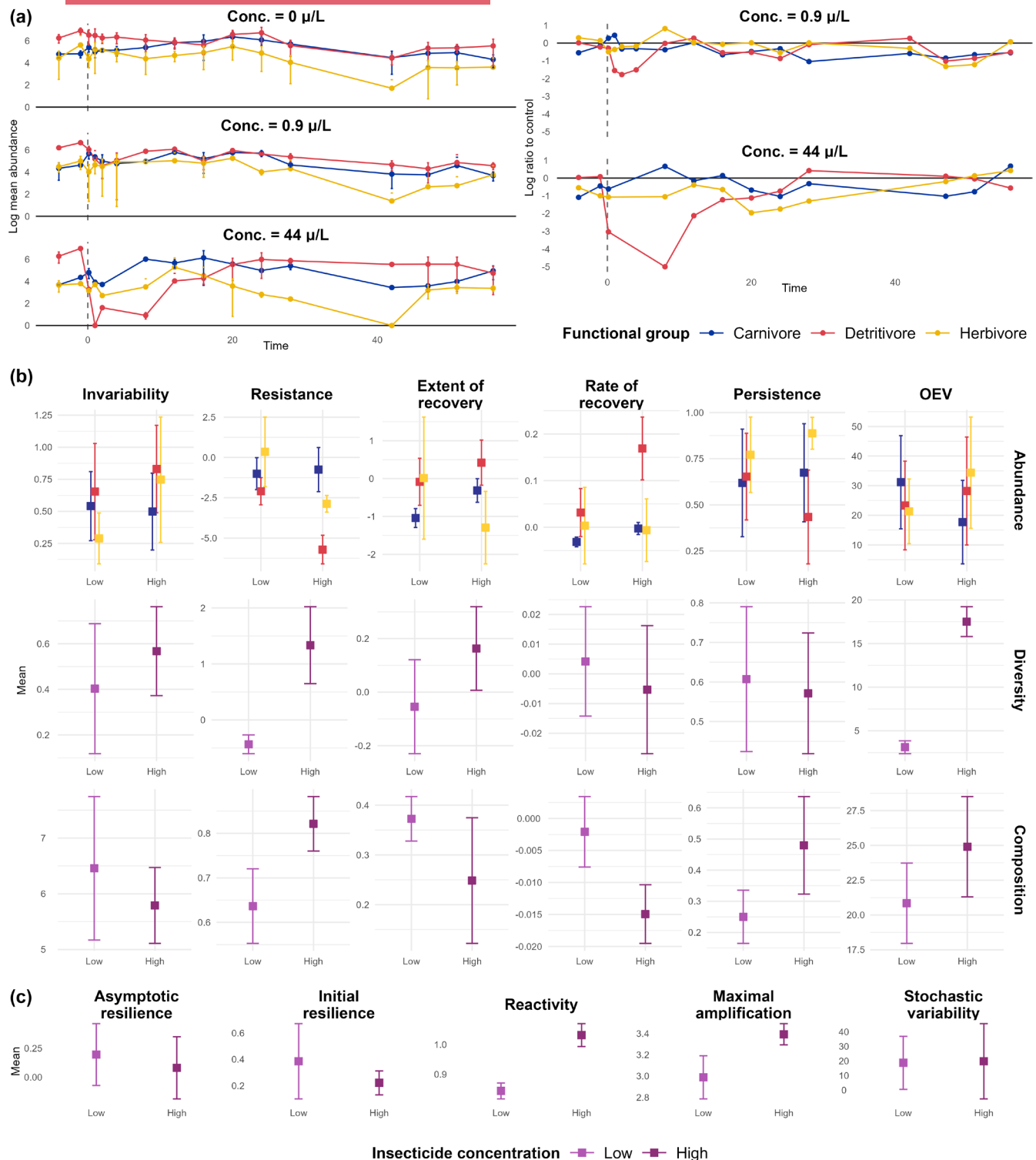


FIGURE 5 Example of *estar* application to quantify the stability of three macroinvertebrate functional groups (carnivores, detritivores and herbivores) to two concentrations of the chlorpyrifos insecticide (0.9 and 44 µg/L, 'low' and 'high', respectively). (a) Log of mean abundance (across replicates) and log-response ratio (to control), over the 60 weeks of the experiment. The insecticide was applied on week 0 (vertical grey dashed line). (b) Functional stability metrics. Values calculated using three state variables: Population abundances ('Abundance' row - caption on the right), Shannon–Wiener index ('Diversity') and community dissimilarity matrices ('Composition'). Shannon–Wiener index and community dissimilarity were calculated using the species in the whole community, that is, not split per functional group. (c) Jacobian metrics. Values shown were averaged over replicates, with error bars indicating the SD.

TABLE 4 Time period and response that we used for the calculation of each functional metric. All functional metrics were calculated for eight pairs of disturbance/control (baseline) replicates, resulting from the combination of four control replicates with two replicates at each insecticide concentration. Jacobian metrics were calculated for each treatment replicate using community data across the entire experimental period.

Functional metrics	Time period	Response
Invariability	From t_{d+1} until t_{end}	Coefficient of variation of the log response ratio
Resistance	Maximal absolute value from t_{d+1} until t_{rec} (week 28)	Log response ratio
Rate of recovery	From t_{d+1} until t_{rec}	Log response ratio
Extent of recovery	At t_{rec}	Log response ratio
Persistence	From t_{d+1} until t_{end}	Abundance or dissimilarity
Overall ecological vulnerability	From t_{d+1} until t_{end}	Log response ratio

hitherto mainly calculated by theoreticians. Thus, *estar* closes the gap between empiricists and theoreticians in stability research and facilitates cross-system comparisons of stability, furthering our understanding of how stable systems are across organisational levels, spatiotemporal scales and environmental conditions.

AUTHOR CONTRIBUTIONS

Viktoriia Radchuk and Ludmilla Figueiredo conceptualised the study and led the design of the package; Paul van den Brink provided the data; Ludmilla Figueiredo and Viktoriia Radchuk analysed the data; Ludmilla Figueiredo, Cédric Scherer and Viktoriia Radchuk wrote the package; Ludmilla Figueiredo led the writing of the manuscript. All authors contributed critically to the package design and drafts and gave final approval for publication.

ACKNOWLEDGEMENTS

We thank the members of the Department of Ecological Dynamics of the Leibniz Institute for Zoo and Wildlife Research (IZW) for fruitful discussions during the conception phase of this project. V.R., S.K.-S. and C.S. are associated with the DFG-funded research training group 'BioMove' (RTG 2118-1) and acknowledge its financial support. S.K. was supported by the Alexander von Humboldt Foundation. Open Access funding enabled and organized by Projekt DEAL.

FUNDING INFORMATION

C.S., S.K.-S. and V.R. were associated with the DFG-funded research training group 'BioMove' (RTG 2118-1) and acknowledge its financial support. S.K. was supported by the Alexander von Humboldt Foundation.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210x.70265>.

DATA AVAILABILITY STATEMENT

All data used are available in the package *estar*, available from CRAN via (<https://doi.org/10.32614/CRAN.package.estar>;

Figueiredo et al., 2025). Further code used in the example is available on Zenodo (<https://doi.org/10.5281/zenodo.18388784>; Figueiredo & Radchuk, 2026).

ORCID

Ludmilla Figueiredo  <https://orcid.org/0000-0001-8217-7800>

Stephanie Kramer-Schadt  <https://orcid.org/0000-0002-9269-4446>

Viktoriia Radchuk  <https://orcid.org/0000-0003-3072-0095>

REFERENCES

- Arnoldi, J.-F., Bideault, A., Loreau, M., & Haegeman, B. (2018). How ecosystems recover from pulse perturbations: A theory of short- to long-term responses. *Journal of Theoretical Biology*, 436, 79–92. <https://doi.org/10.1016/j.jtbi.2017.10.003>
- Arnoldi, J.-F., Loreau, M., & Haegeman, B. (2016). Resilience, reactivity and variability: A mathematical comparison of ecological stability measures. *Journal of Theoretical Biology*, 389, 47–59. <https://doi.org/10.1016/j.jtbi.2015.10.012>
- Capdevila, P., Stott, I., Beger, M., & Salguero-Gómez, R. (2020). Towards a comparative framework of demographic resilience. *Trends in Ecology & Evolution*, 35(9), 776–786. <https://doi.org/10.1016/j.tree.2020.05.001>
- Carrara, F., Giometto, A., Seymour, M., Rinaldo, A., & Altermatt, F. (2015a). Experimental evidence for strong stabilizing forces at high functional diversity of aquatic microbial communities. *Ecology*, 96(5), 1340–1350. <https://doi.org/10.1890/14-1324.1>
- Carrara, F., Giometto, A., Seymour, M., Rinaldo, A., & Altermatt, F. (2015b). Inferring species interactions in ecological communities: A comparison of methods at different levels of complexity. *Methods in Ecology and Evolution*, 6(8), 895–906. <https://doi.org/10.1111/2041-210X.12363>
- Domínguez-García, V., Dakos, V., & Kéfi, S. (2019). Unveiling dimensions of stability in complex ecological networks. *Proceedings of the National Academy of Sciences*, 116(51), 25714–25720. <https://doi.org/10.1073/pnas.1904470116>
- Donohue, I., Hillebrand, H., Montoya, J. M., Petchey, O. L., Pimm, S. L., Fowler, M. S., Healy, K., Jackson, A. L., Lurgi, M., McClean, D., O'Connor, N. E., O'Gorman, E. J., & Yang, Q. (2016). Navigating the complexity of ecological stability. *Ecology Letters*, 19(9), 1172–1185. <https://doi.org/10.1111/ele.12648>
- Donohue, I., Petchey, O. L., Montoya, J. M., Jackson, A. L., McNally, L., Viana, M., Healy, K., Lurgi, M., O'Connor, N. E., & Emmerson, M. C. (2013). On the dimensionality of ecological stability. *Ecology Letters*, 16(4), 421–429. <https://doi.org/10.1111/ele.12086>
- Downing, A. L., Jackson, C., Plunkett, C., Lockhart, J. A., Schlatter, S. M., & Leibold, M. A. (2020). Temporal stability vs. community matrix

- measures of stability and the role of weak interactions. *Ecology Letters*, 23(10), 1468–1478. <https://doi.org/10.1111/ele.13538>
- Figueiredo, L., & Radchuk, V. (2026). Notebook with the analysis of stability of an aquatic macroinvertebrate community using the *estar* package. Zenodo <https://doi.org/10.5281/zenodo.18388784>
- Figueiredo, L., Radchuk, V., & Scherer, C. (2025). *estar: Ecological Stability Metrics* (Version 1.0–0) [Computer software]. <https://doi.org/10.32614/CRAN.package.estar>
- Grimm, V., & Wissel, C. (1997). Babel, or the ecological stability discussions: An inventory and analysis of terminology and a guide for avoiding confusion. *Oecologia*, 109(3), 323–334. <https://doi.org/10.1007/s004420050090>
- Hallett, L., Avolio, M. L., Carroll, I. T., Jones, S. K., MacDonald, A. A. M., Flynn, D. F. B., Slaughter, P., Ripplinger, J., Collins, S. L., Gries, C., & Jones, M. B. (2020). *codyn: Community Dynamics Metrics* [Computer software]. <https://doi.org/10.5063/F1N877Z6>
- Hampton, S. E., Holmes, E. E., Scheef, L. P., Scheuerell, M. D., Katz, S. L., Pendleton, D. E., & Ward, E. J. (2013). Quantifying effects of abiotic and biotic drivers on community dynamics with multivariate autoregressive (MAR) models. *Ecology*, 94(12), 2663–2669. <https://doi.org/10.1890/13-0996.1>
- Hillebrand, H., & Kunze, C. (2020). Meta-analysis on pulse disturbances reveals differences in functional and compositional recovery across ecosystems. *Ecology Letters*, 23(3), 575–585. <https://doi.org/10.1111/ele.13457>
- Hillebrand, H., Langenheder, S., Lebrecht, K., Lindström, E., Östman, Ö., & Striebel, M. (2018). Decomposing multiple dimensions of stability in global change experiments. *Ecology Letters*, 21(1), 21–30. <https://doi.org/10.1111/ele.12867>
- Holling, C. S. (1973). Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4(1), 1–23. <https://doi.org/10.1146/annurev.es.04.110173.000245>
- Holmes, E. E., Scheuerell, M. D., & Ward, E. J. (2024). Analysis of multivariate time-series using the MARSS package. Version 3.11.9. <https://doi.org/10.5281/zenodo.5781847>
- Holmes, E. E., Ward, E. J., Scheuerell, M. D., & Wills, K. (2024). MARSS: *Multivariate Autoregressive State-Space Modeling* (Version 3.11.9) [Computer software]. <https://doi.org/10.32614/CRAN.package.MARSS>
- Holmes, E. E., Ward, E. J., & Wills, K. (2012). MARSS: Multivariate autoregressive state-space models for analyzing time-series data. *The R Journal*, 4(1), 30. <https://doi.org/10.32614/RJ-2012-002>
- Ingrisch, J., & Bahn, M. (2018). Towards a comparable quantification of resilience. *Trends in Ecology & Evolution*, 33(4), 251–259. <https://doi.org/10.1016/j.tree.2018.01.013>
- Ives, A. R., Dennis, B., Cottingham, K. L., & Carpenter, S. R. (2003). Estimating community stability and ecological interactions from time-series data. *Ecological Monographs*, 73(2), 301–330. [https://doi.org/10.1890/0012-9615\(2003\)073%5B0301:ECSAEI%5D2.0.CO;2](https://doi.org/10.1890/0012-9615(2003)073%5B0301:ECSAEI%5D2.0.CO;2)
- Medeiros, L. P., Song, C., & Saavedra, S. (2021). Merging dynamical and structural indicators to measure resilience in multispecies systems. *Journal of Animal Ecology*, 90(9), 2027–2040. <https://doi.org/10.1111/1365-2656.13421>
- Neubert, M. G., & Caswell, H. (1997). Alternatives to resilience for measuring the responses of ecological systems to perturbations. *Ecology*, 78(3), 653–665. [https://doi.org/10.1890/0012-9658\(1997\)078%5B0653:ATRFMT%5D2.0.CO;2](https://doi.org/10.1890/0012-9658(1997)078%5B0653:ATRFMT%5D2.0.CO;2)
- Oksanen, J., Simpson, G. L., Blanchet, F. G., Kindt, R., Legendre, P., Minchin, P. R., O'Hara, R. B., Solymos, P., Stevens, M. H. H., Szoecs, E., Wagner, H., Barbour, M., Bedward, M., Bolker, B., Borcard, D., Carvalho, G., Chirico, M., Caceres, M. D., Durand, S., ... Weedon, J. (2024). *vegan: Community Ecology Package* (Version 2.6–8) [Computer software]. <https://doi.org/10.32614/CRAN.package.vegan>
- Pimm, S. L. (1984). The complexity and stability of ecosystems. *Nature*, 307(5949), 5949. <https://doi.org/10.1038/307321a0>
- Radchuk, V., Laender, F. D., Cabral, J. S., Boulangeat, I., Crawford, M., Bohn, F., Raedt, J. D., Scherer, C., Svenning, J.-C., Thonicke, K., Schurr, F. M., Grimm, V., & Kramer-Schadt, S. (2019). The dimensionality of stability depends on disturbance type. *Ecology Letters*, 22(4), 674–684. <https://doi.org/10.1111/ele.13226>
- Ryo, M., Aguilar-Trigueros, C. A., Pinek, L., Muller, L. A. H., & Rillig, M. C. (2019). Basic principles of temporal dynamics. *Trends in Ecology & Evolution*, 34(8), 723–733. <https://doi.org/10.1016/j.tree.2019.03.007>
- Scheef, L. P., & Holmes, E. E. (2023). *MAR1: Multivariate Autoregressive Modeling for Analysis of Community Time-Series Data* [Computer software]. <https://cran.r-project.org/package=MAR1>
- Stott, I., Hodgson, D., & Townley, S. (2021). *popdemo: Demographic Modelling Using Projection Matrices* (Version 1.3–2) [Computer software]. <https://doi.org/10.32614/CRAN.package.popdemo>
- Urrutia-Cordero, P., Langenheder, S., Striebel, M., Angeler, D. G., Bertilsson, S., Eklöv, P., Hansson, L.-A., Kelpsiene, E., Laudon, H., Lundgren, M., Parkefeld, L., Donohue, I., & Hillebrand, H. (2022). Integrating multiple dimensions of ecological stability into a vulnerability framework. *Journal of Ecology*, 110(2), 374–386. <https://doi.org/10.1111/1365-2745.13804>
- van den Brink, P. J., Van Wijngaarden, R. P. A., Lucassen, W. G. H., Brock, T. C. M., & Leeuwangh, P. (1996). Effects of the insecticide dursban® 4E (active ingredient chlorpyrifos) in outdoor experimental ditches: II. Invertebrate community responses and recovery. *Environmental Toxicology and Chemistry*, 15(7), 1143–1153. <https://doi.org/10.1002/etc.5620150719>
- Van Meerbeek, K., Jucker, T., & Svenning, J.-C. (2021). Unifying the concepts of stability and resilience in ecology. *Journal of Ecology*, 109(9), 3114–3132. <https://doi.org/10.1111/1365-2745.13651>
- White, L., O'Connor, N. E., Yang, Q., Emmerson, M. C., & Donohue, I. (2020). Individual species provide multifaceted contributions to the stability of ecosystems. *Nature Ecology & Evolution*, 4(12), 1594–1601. <https://doi.org/10.1038/s41559-020-01315-w>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Data S1: Summarized values (Tables S1–S4), and detailed sub-figures used to generate Figure 5 of the main text.

How to cite this article: Figueiredo, L., Scherer, C., Kramer-Schadt, S., Cabral, J. S., Kéfi, S., Van den Brink, P. J., & Radchuk, V. (2026). *estar*: An R package to measure ecological stability. *Methods in Ecology and Evolution*, 17, 1090–1100. <https://doi.org/10.1111/2041-210x.70265>